

Degree: B.Tech.(ECE-AIML), Semester: IIIrd
MID-SEMESTER EXAMINATION, September, 2024

Course Title: **Microelectronics Circuits and Applications**
Course Code: **EAEPC-07**

Duration: **1:30 Hours**Max. Marks: **15**

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1a	Draw the neat output characteristic of an <i>nnp</i> BJT, explain and exhibit the effect of base-width modulation thereon.	1½	CO1
1b	An <i>nnp</i> transistor's J_{CB} is reverse biased and emitter is left open circuited. Utilising Ebers-Moll model description of the BJT determine the mode of its operation and the values of I_C , I_B and V_{EB} .	1½	CO1
2a	Derive the expression for the transconductance of the MOSFET in saturation region and draw the low-frequency small-signal model.	1½	CO1
2b	Derive the A_v , A_i , R_i , R_o of the CE and CB configurations and compare their typical magnitudes.	1½	CO1
3a	With neat circuit diagrams and necessary derivations, discuss BJT and MOSFET based simple current mirrors. Compare their features too.	1½	CO2
3b	Discuss the miller effect and derive the f_H of the common-emitter BJT configuration. (Use the hybrid-Pi model for high frequency)	1½	CO2
4a	Design the circuit given in Fig.4a to establish the $I_{CQ}=0.2\text{mA}$ and $V_C=0.5\text{V}$. The transistor exhibits v_{BEQ} of 0.8V at $i_C=1\text{mA}$ and $\beta=100$.	1½	CO2
4b	For the circuit given in Fig.4b calculate the DC node voltages V_1 and V_2 for $V_T=1\text{V}$ and $k'(W/L)=4\text{mA/V}^2$ and $\lambda=0$.	1½	CO3
5a	Calculate A_v , R_i and R_o for the circuit given in Fig.5a	1½	CO3
5b	Derive the transconductance (g_m) of the BJT differential pair and plot the v_{dm} versus $I_{diffout}$ characteristic of the same.	1½	CO3

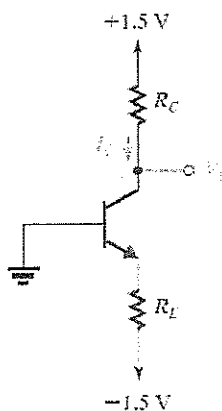


Fig.4a

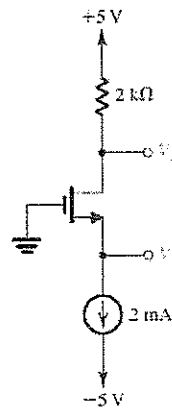


Fig.4b

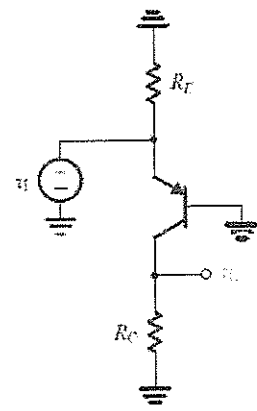


Fig.5a